

GFM Envelope Generation — Producer.ini (Fusion)

DyCoV Tool — `dycov generateEnvelopes` command | File: `examples/GFM/Fusion/Producer.ini`

1. Commands to Run

Step 1 — Open terminal and go to the project folder

```
cd <project-folder>/dyn-grid-compliance-verification
```

Step 2 — Activate the Python virtual environment

```
source .venv/bin/activate
```

Step 3 — Run the envelope generation command

```
dycov generateEnvelopes -i "examples/GFM/Fusion/Producer.ini" -o  
"examples/GFM/Fusion/Results"
```

Flag meaning:

- `-i` — path to the Producer INI file with the plant's parameters (D, H, Xeff, Snom, p_max, etc.)
- `-o` — output folder where all generated curves (CSV / PNG / HTML) will be saved

If the output folder already has results from a previous run, the tool will ask:

```
Do you want to overwrite the results directory? (y/[N])
```

Type `y` and press Enter to regenerate.

2. What Producer.ini Actually Controls

The Producer.ini file supplies the electrical/mechanical parameters of the plant. It does **not** define which grid events get tested — those 4 event types are fixed inside the DyCoV tool itself.

Parameter (from Producer.ini)	Meaning
p_max_injection / p_min_injection	Active power injection limits (MW)
u_nom	Nominal voltage at the PDR bus (kV)
q_max / q_min	Reactive power limits (MVar)
Snom	Nominal apparent power (MVA)
D_Overdamped / D_Underdamped	Damping factor for each damping regime
H_Overdamped / H_Underdamped	Inertia constant for each damping regime
Xeff	Effective reactance (pu)

3. Which Envelopes Get Generated

DyCoV always runs **4 fixed GFM benchmark templates** (defined in the tool's PCS templates, same for every producer) and combines them with the D/H/Xeff values above:

PCS (Benchmark)	Grid Event Tested	Event Variants	Operating Conditions (OC) each	Curve Sets Generated
PCS_RTE-IGFM1	Voltage Angle Step	S_VolAngStep1 – 4	5	20
PCS_RTE-IGFM2	Voltage Amplitude Step	S_VolAmpStep1 – 4	7	28
PCS_RTE-IGFM3	SCR Jump (Up / Down)	S_SCRup1–2, S_SCRdown1–2	5	20
PCS_RTE-IGFM4	RoCoF (Rate of Change of Frequency)	S_Rocof1 – 8	5	40
Total				108 curve sets

Each curve set = 1 `.csv` (raw data) + 1 `.png` (static plot) + 1 `.html` (interactive plot).

Output location

```
examples/GFM/Fusion/Results/  
├── PCS_RTE-IGFM1/S_VolAngStep[1–4]/OC[1–5]/...  
├── PCS_RTE-IGFM2/S_VolAmpStep[1–4]/OC[1–7]/...  
├── PCS_RTE-IGFM3/S_SCRup[1–2] , S_SCRdown[1–2]/OC[1–5]/...  
└── PCS_RTE-IGFM4/S_Rocof[1–8]/OC[1–5]/...
```

Note on the Overdamped case: With the current values ($D_{\text{Overdamped}} = 152$, $H_{\text{Overdamped}} = 2$), the tool's own $\pm 15\%$ parameter-tolerance check found that one variation crosses into the underdamped region (damping ratio ε drops to 0.96, below the $\varepsilon \geq 1$ overdamped threshold). The affected plots are still generated but carry a "Disclaimer: Inconsistent damping" note. Increasing the margin between $D_{\text{Overdamped}}/H_{\text{Overdamped}}$ and the $\varepsilon = 1$ boundary would remove this disclaimer.